

8. Which of the following is a ternary operator?

- A) &&
C) ?:

- B) >>=
D) ->

9. What is the result of the following code?

```
int x=0;
```

```
switch(x)
```

```
{
```

```
case 1: printf( "One" );
```

```
case 0: printf( "Zero" );
```

```
case 2: printf( "Hello World" );
```

```
}
```

- A) One
C) Hello World

- B) Zero
D) ZeroHello World

10. A "switch" statement is used to

- A) Switch between functions in a program
C) To choose from multiple possibilities which may arise due to different values of a single variable

- B) Switch from one variable to another variable
D) All of above

11. Consider following program fragment

```
if (a > b)
```

```
printf("a> b") ;
```

```
else
```

```
printf ("else part") ;
```

```
printf ("a< = b") a <= b
```

```
will be printed if
```

- A) a>b
C) a=b

- B) a < b
D) All of these

12. What is the maximum number of dimensions an array in C may have?

- A) Two
C) Twenty

- B) Eight
D) Theoretically no limit. The only practical limits are memory size and compilers.

13. If S is an array of 80 characters, then the value assigned to S through the statement scanf("%s",S) with input 12345 would be

- A) "12345"
C) S is an illegal name for string

- B) nothing since 12345 is an integer
D) %s cannot be used for reading in values of S

14. Size of the array need not be specified, when

- A) Initialization is a part of definition
C) It is a formal parameter

- B) It is a declaration
D) All of these

15. In the following 'C' code, in which order the functions would be called ?

```
a = ( f1(23,14) * f2 (12/14)) + f3 () ;
```

```
A) f1,f2,f3
```

```
B) f3,f2,f1
```

```
C) The order may vary from compiler to compiler
```

```
D) None of these
```

16. What error would the following function give on compilation ?

```
f(int a,int b)
```

```
{
```

```
int a;
```

```
a = 20;
```

```
return a;
```

```
}
```

- A) Missing parentheses in return statement
C) Redeclaration of a

- B) Function should be define as int f(int a,int b)
D) No error

17. Which function is used to close opened files?
 A) Open B) fopen
 C) Close D) fclose
18. Which of the following is used to append a file?
 A) r B) w
 C) x D) a
19. When `strcpy(str1, str2)` is executed,
 A) str1 is stored in str2 B) str2 is stored in str1
 C) copied string is stored in virtual memory D) None of the above
20. Which of the following is not returned by `strcmp`.
 A) 1 B) value >0
 C) value <0 D) infinity

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	List the different Computer generation types with 2 significant characteristics for each type. Write any 4 differences between primary and secondary memory.	07	L*2	1	1
b)	Define Keywords in C and list any 4 keywords. Identify the valid identifiers among the following. State the reason if invalid. i. SuM@234 ii. 12Re_TEST iii. Hello_World! iv. sum_of_3_numbers	05 04	L1 L3	1 2	1 2
c)	Construct a C Program to swap two numbers.				
2. a)	Define the terms Algorithm and Flowchart. Write the algorithm and flowchart to find area of triangle where base and height of triangle have to be taken as input, and $\text{Area} = 1/2 \times \text{base} \times \text{height}$.	06	L3	1	2
b)	Differentiate between formatted and unformatted input/output functions in C. Explain any 2 unformatted input/output functions in C with relevant syntax.	06	L2	2	1
c)	What is type conversion in C? Explain explicit type conversion with relevant programming example.	04	L2	2	1
3. a)	Solve the following expression in C. i. $x = a - b / (3 + c) * (2 - 1)$ where $a=9$, $b=12$ and $c=3$ ii. $a / b \leq c - d + a \% 10 - b == d \geq e != b$ where $a=100$, $b=20$, $c=10$, $d=5$, $e=1$	07	L3	2	2
b)	List the fundamental datatypes available in C and explain any 2 datatypes and their usage in C Programs.	05	L2	1	1
c)	Illustrate usage of ternary operator by finding largest of two numbers.	04	L2	2	1
Unit – II					
4. a)	Write a C program to calculate sum of n natural numbers using while loop.	04	L3	3	2
b)	With example, explain with syntax how to declare 1-dimensional and different ways to initialize 1-dimensional arrays in C.	06	L2	4	1
c)	Describe with syntax and flowchart, else-if ladder in C.	06	L2	3	1
5. a)	Differentiate between entry-controlled loop and exit controlled loop.	04	L2	3	1
b)	Determine the various elements of User defined functions with an example.	06	L2	4	1
c)	Construct a C program to transpose a matrix of order M x N and find the trace of the resultant matrix.	06	L3	4	2

6. a) Develop a C program, to implement a bubble sort technique using functions to sort given N integers in ascending order. Display the input array and sorted array with suitable headings.
- b) Explain continue and goto statement with example and their use.
- c) Compare and contrast Call by Value and Call by Reference.

06	L3	4	2
06	L2	3	1
04	L2	4	1

Unit – III

7. a) Explain the following string handling functions with their syntax and give example for each.
- i. strcmp()
- ii. strncat()
- iii. strlen()
- iv. strcpy()
- b) Create a structure called Employee with the fields: Name, Designation, Salary and Experience in years. Input the details of 2 employees and display the same with suitable labels.
- c) Define a pointer. Explain the benefits of using pointers in C.
8. a) Write a C program to read N integers into an array A and find the sum of elements using pointers.
- b) What do you mean by string? With example explain string representation.
- c) Write the general syntax of following File handling functions with their usage.
- fopen(), fprintf(), fscanf(), getc().

06	L2	5	1
06	L3	5	2
04	L2	5	1
06	L3	5	2
04	L2	5	1
06	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
